INTERNATIONAL MONETARY FUND

CABO VERDE

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STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION — JOINT IDA/IMF DEBT SUSTAINABILITY ANALYSIS

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While Cabo Verde's debt remains sustainable, it is subject to significant vulnerabilities. Based on the authorities' budgetary projections ("baseline"), Cabo Verde's external debt risk distress rating is "moderate". However, total public debt is expected to reach considerably higher levels over the medium term than projected in the previous Debt Sustainability Analysis (DSA). This is a dangerous trend, particularly in light of a weaker macroeconomic outlook and contingent financial liabilities related to state-owned enterprises. Reducing the risk of debt distress depends critically on accelerating fiscal consolidation and advancing structural reforms to boost growth and enhance the efficiency of the public sector.

BACKGROUND

Over the last five years, Cabo Verde's public debt has increased significantly, reflecting a sharp decline in revenue and grants, as well as the ramping up of public investment. While public and publicly guaranteed debt remains sustainable, it is expected to reach considerably higher levels over the medium term than projected in the previous DSA, and is subject to significant vulnerabilities. This is a dangerous trend, particularly in light of a weaker macroeconomic outlook and contingent financial liabilities related to the debt of state-owned enterprises. Given these risks, efforts to significantly accelerate fiscal consolidation, combined with structural reforms that boost productivity and growth, are essential to improve the debt outlook.

1. This DSA reflects official debt stock data for end-2012, and additional information available as of April 2014. The fiscal outturn of 2013 and staff's fiscal projections for 2014–17 (Table 3a in the Staff Report) are used to project debt-creating flows. The debt data include central government external and domestic debt, and external debt contracted by the central government on behalf of state-owned enterprises (also referred to as "onlending"). The data do not include debt contracted directly by state-owned enterprises and local governments that carry a central-government guarantee; at end-2012, this debt was estimated at about 4 percent of GDP. Estimates for such publicly-guaranteed debt for 2013, and projections for 2014 and beyond, were not available at the time of this assessment.

2. There is little reliable data on external private debt. To estimate this debt, information from the balance of payments on bank and non-bank liabilities to non-residents is used. On this basis, private external debt was estimated around 36 percent of GDP at end-2013. The authorities are compiling information about private sector external debt, but there is a need for a more systematic monitoring of actual disbursements and repayment flows.

3. Total public debt in Cabo Verde continued to increase in 2013 as a result of the Public Investment Program (PIP). The stock of public debt reached 90 percent of GDP in 2012, and is estimated to have reached 98 percent of GDP at end-2013. The increase in public debt reflects the still-high financing needs owing to increases in capital spending, and falling tax revenue as a share of GDP.

Text Table 1. Cabo Verde: Stock of Total Public Debt at End-Year, 2004–13 [1]

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013 2/
					(Percent of GDP)					
External debt	50.1	49.6	46.0	40.3	39.2	43.3	51.3	56.1	67.5	74.0
Domestic debt	33.7	35.7	31.7	24.7	19.1	21.3	21.7	21.6	22.6	23.6
Total public debt	83.7	85.3	77.7	65.0	58.2	64.6	73.0	77.8	90.2	97.6

Sources: Cabo Verdean authorities and IMF staff calculations.

1/ GDP data for 2013 are IMF staff estimates.
2/ The public debt level in 2013 is an estimate based on end-2012 debt stock and debt-creating flows in 2013.

4. While external public debt is high, it is overwhelmingly concessional. Multilateral institutions, in particular the World Bank Group and the African Development Bank, are the main external creditors. Cabo Verde's external public debt has a long maturity profile and low average interest rates (Text Table 2).

Text Table 2. Cabo Verde: External Debt Profile by Type of Creditor, 2012

	Percent of total external debt	Average grace period	Average amortization period	Average interest rate
Multilateral	52	9	34	1.0%
Bilateral	24	7	17	1.2%
Commercial	24	10	20	1.7%

Sources: Cabo Verdean authorities and IMF staff estimates.

5. Cabo Verde's domestic public debt remains at relatively low levels. The government's ability to finance the PIP through concessional external loans has helped keep domestic debt low (24 percent of GDP at end-2013). Moreover, there is a legal limit of 3 percent of GDP on domestic financing of the deficit per year.[1] At end-2012 the National Pension Fund held about 44 percent of domestic debt, and the rest was held by the banking system. Treasury bonds make up about 94 percent of domestic debt. The average maturity of domestic debt at end-2013 was six years and the average interest rate was 5.4 percent.

6. Cabo Verde is a strong policy performer for the purpose of determining the debt burden thresholds under the Debt Sustainability Framework (DSF). Cabo Verde's rating on the World Bank's Country Policy and Institutional Assessment (CPIA) averaged 4 (on a scale of 1 to 6) during 2010–12, making it a strong policy performer. Based on its 2012 CPIA score, Cabo Verde ranks first among IDA-recipient countries in the sub-Saharan African (SSA) region, and third among all IDA countries. The corresponding external public debt burden thresholds are shown in Text Table 3.

[1] In addition to the legal limit on domestic deficit financing, the government has also set the policy objective of keeping domestic debt below 25 percent of GDP.

Text Table 3. External Public Debt Thresholds for Strong Policy Performers under the Debt Sustainability Framework

Present value of external debt, percent of:	
GDP	50
Exports	200
Revenue	300
External debt service in percent of:	
Exports	25
Revenue	22

BASELINE SCENARIO UNDERLYING THE DEBT SUSTAINABILITY ANALYSIS

7. The assumptions underlying the current DSA differ in a number of ways from those used in the 2012 Article IV Consultation (Text Table 4 and Box 1). While the current baseline scenario assumes a rebound of economic activity, growth projections in the medium and long term are about one percentage point lower than in the previous DSA. Growth is also estimated to have been considerably weaker in 2012 and 2013. The baseline scenario assumes a slower path of fiscal consolidation, given the delayed economic recovery and higher current account deficit, reflecting higher imports owing to a slower phasing out of the PIP.

Text Table 4. Cabo Verde: Assumptions for Key Economic Indicators, 2012–17 1/
(Percent of GDP)

	2012	2013	2014	2015	2016	2017
Real GDP growth						
Current DSA	1.2	0.5	3.0	3.5	4.0	4.0
2012 Article IV DSA	4.3	4.1	4.5	4.7	5.0	5.0
Fiscal balance (including grants)						
Current DSA	-9.9	-7.8	-7.4	-6.5	-5.7	-4.5
2012 Article IV DSA	-7.5	-7.6	-5.8	-2.2	0.2	2.7
Fiscal balance (including onlending)						
Current DSA	-13.6	-12.9	-11.8	-10.5	-9.4	-6.6
2012 Article IV DSA	-11.2	-12.2	-10.1	-5.7	-0.2	2.3
Current account deficit (including grants)						
Current DSA	-11.3	-4.5	-9.8	-10.7	-11.3	-10.8
2012 Article IV DSA	-11.1	-13.2	-11.4	-8.3	-6.1	-4.3

Sources: Cabo Verdean authorities and IMF staff estimates.

1/ GDP data for 2013 are IMF staff estimates. All data for 2014–17 are IMF staff projections.

Box 1. Cabo Verde: Macroeconomic Assumptions of the Baseline Scenario, 2013–33

Real GDP growth is expected to pick up after the slowdown in 2012-13, and settle at 4 percent per year in the long term. Growth assumptions are based on continued good performance in the tourism sector, better conditions in the euro area, resumption of private credit growth, some product diversification into areas like agriculture and fisheries, and an increase in productivity owing to payoffs from the Public Investment Program and from structural reforms.

Fiscal policy. In the medium term (2013–17), the fiscal deficit and overall financing needs are expected to decline by 2.8 and 6.3 percentage points of GDP, respectively, corresponding to the fiscal consolidation outlined in the baseline scenario, as set out in paragraph 23 in the Staff Report. Fiscal consolidation is expected to continue in the long run, with the government continuing to reduce its financing needs as the concessional borrowing window closes. In the long run (2018-33), with fiscal consolidation complete and onlending coming to an end, financing needs are projected to stay below 2.5 percent of GDP per year.

The non-interest current account deficit is projected to widen in 2014 owing to the expected increase in economic activity, public investment and FDI (which will drive up imports) and decline gradually afterwards, as the Public Investment Program winds down.

Consumer price inflation is projected to remain below 3 percent.

Financing. The concessionality of new external loans will decline significantly starting from 2019, and the baseline assumes a slightly more accelerated move towards non-concessional financing than the previous DSA. In addition, the baseline assumes that domestic financing will remain below 3 percent of GDP, in compliance with the limit set in the Budget Law.

DEBT SUSTAINABILITY ANALYSIS

E. External Public Debt

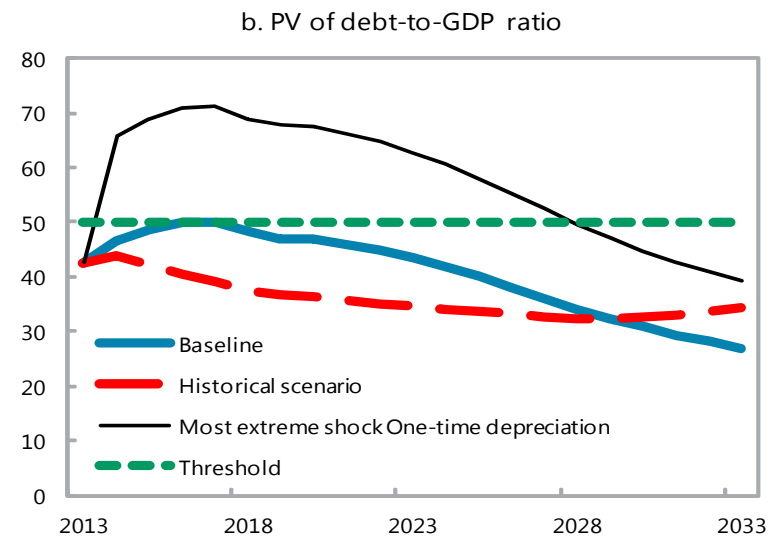
8. Under the baseline assumptions, all external debt and debt-service ratios remain below the policy-dependent thresholds throughout the projection period (Figure 1). The present value (PV) of public and publicly guaranteed (PPG) external debt is expected to peak at 49 percent of GDP in 2016, and then gradually decrease to 28 percent of GDP in 2033 (Figure 1b, Table 1a). The large increase in the PV of PPG external debt in 2013-14 is due to high public investment in these years.

9. Cabo Verde's external debt sustainability is most sensitive to a one-time depreciation shock. Under a bound test that entails a 30 percent nominal depreciation shock in 2014, the PV of the debt-to-GDP ratio would exceed the threshold in the year 2014 and remain above it for more than a decade (Figure 1b; Table 1b, Scenario B6). The PV of the debt-to-exports ratio would also breach the threshold under a hardening of financial terms, an export shock, and a GDP deflator shock (Table 1b, scenario A2, B2, and B3, respectively).

10. External debt sustainability is also vulnerable to contingent liabilities associated with the debt of state-owned enterprises (SOEs). At end-2012 (the latest information available), SOEs' debt that did *not* carry a government guarantee amounted to 12 percent of GDP. In a shock scenario where the financial situation of SOEs deteriorates to such an extent that the central government

must take on all of this debt, and respecting legal limits on domestic debt financing, the PV of PPG external debt reaches the 50 percent threshold under the baseline (Text Figure 1).[2]

Text Figure 1. Cabo Verde: Public and publicly guaranteed external debt in SOE debt scenario, 2013–33



Sources: Country authorities; and staff estimates and projections.

F. Total Public Debt

11. Total public debt in the baseline scenario peaks at 108 percent of GDP in 2016, and declines gradually over the projection period (Table 2a). The debt outlook is vulnerable to a prolonged economic slowdown; a delay of the Euro zone recovery, which would affect growth by depressing remittances and tourism income; and realization of losses on contingent liabilities associated with SOEs. With regards to remittances, which accounted for roughly 10 percent of GDP in 2013, there are also vulnerabilities related to financial stability. While these flows have remained fairly stable over many years, problems in a systemic bank could cause emigrants to re-assess their financial investment in Cabo Verde and withdraw their funds. Regarding SOEs, while the financial situation of ELECTRA has improved during 2013, TACV continues to be in difficult financial circumstances.

[2] The scenario assumes that (i) the stock of non-guaranteed SOE debt held steady in 2013 at the level recorded for end-2012 (CV$ 18.2 billion) and (ii) the entire stock of this debt is transferred to the central government in 2014. Assuming that the SOE debt at end-2012 had an average original maturity of 7 years, the government would need to refinance this debt over 2014–19. The financing to cover this would come from domestic borrowing (up to the legally binding limit on domestic deficit financing of 3 percent of GDP) and the remainder from external borrowing.

12. Public debt sustainability is sensitive to various alternative scenarios and stress tests (Table 2b). The expansion of public debt is most pronounced under the scenario which keeps the primary balance the same as in 2013. However, this scenario appears rather unlikely, given that the primary balance in 2013 reflected a temporarily high level of public investment. Significantly lower growth (for a prolonged period or the whole projection horizon) would also cause public debt to reach unsustainable levels.

G. Comparison with the Previous Debt Sustainability Analysis

13. While the PV of external debt does not breach the threshold under the baseline scenario—as was the case in the previous DSA—public debt peaks at a higher level. The improved assessment under the baseline is due to the increase in the discount rate from 3 to 5 percent.[3] However, while in the previous DSA public debt peaked at 92 percent of GDP (in 2015), in the current DSA it reaches 108 percent in 2016. This is a significant deterioration in the outlook for public debt, reflecting a combination of weaker economic growth and a slower path of fiscal consolidation.

H. The Authorities' Views

14. The authorities broadly concurred with the results of the DSA, though they saw upside risks to the assessment. In particular, they noted that by improving Cabo Verde's infrastructure, the large-scale public investment program was addressing critical bottlenecks to growth, which would deliver a more favorable economic performance over the medium term. This would bolster domestic revenue (which would reduce the deficit) and also raise growth (which would reduce the debt-to-GDP ratio).

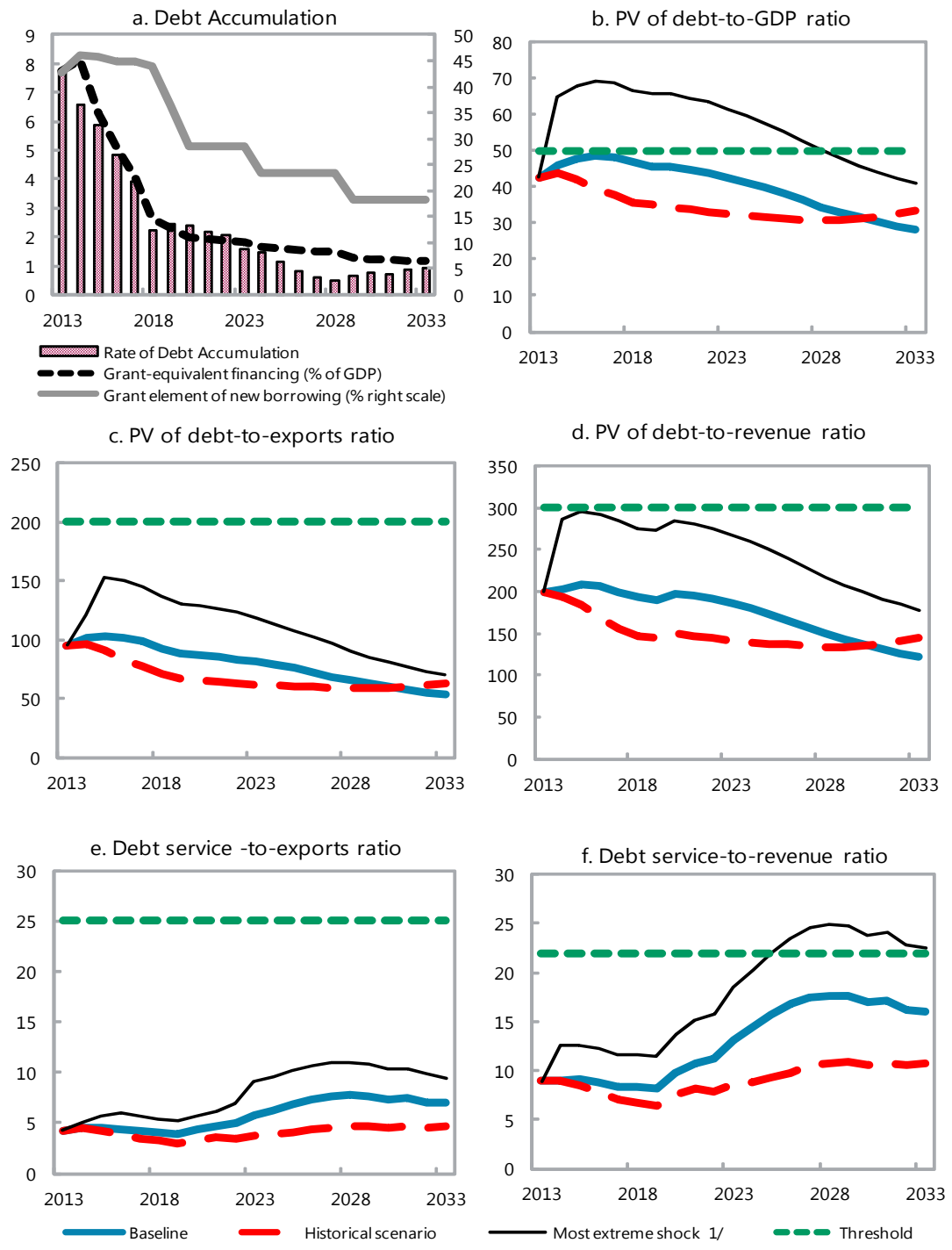
DEBT DISTRESS CLASSIFICATION AND CONCLUSIONS

15. Based on the external debt burden indicators, the current DSA finds that Cabo Verde's debt is sustainable, and that the risk of debt distress is moderate. None of the debt burden thresholds is breached over the 20-year projection period under the baseline scenario. A depreciation shock is the one to which debt sustainability is most sensitive.

16. However, the rapid accumulation of public debt has increased risks to the baseline, and points to elevated vulnerabilities. The degree (and timing) of the impact of the public investment program on growth remains uncertain, prospects for export diversification are challenging, and SOEs remain in financial difficulties. Given these downside risks, further efforts should be made to accelerate fiscal consolidation and reduce public debt faster. In addition, the government should aim to limit its external borrowing to concessional loans.

[3] The discount rate for the IMF's Debt Sustainability Analysis was increased from 3 to 5 percent in 2013 because globally low interest rates were making it difficult to obtain financing terms that met concessionality requirements.

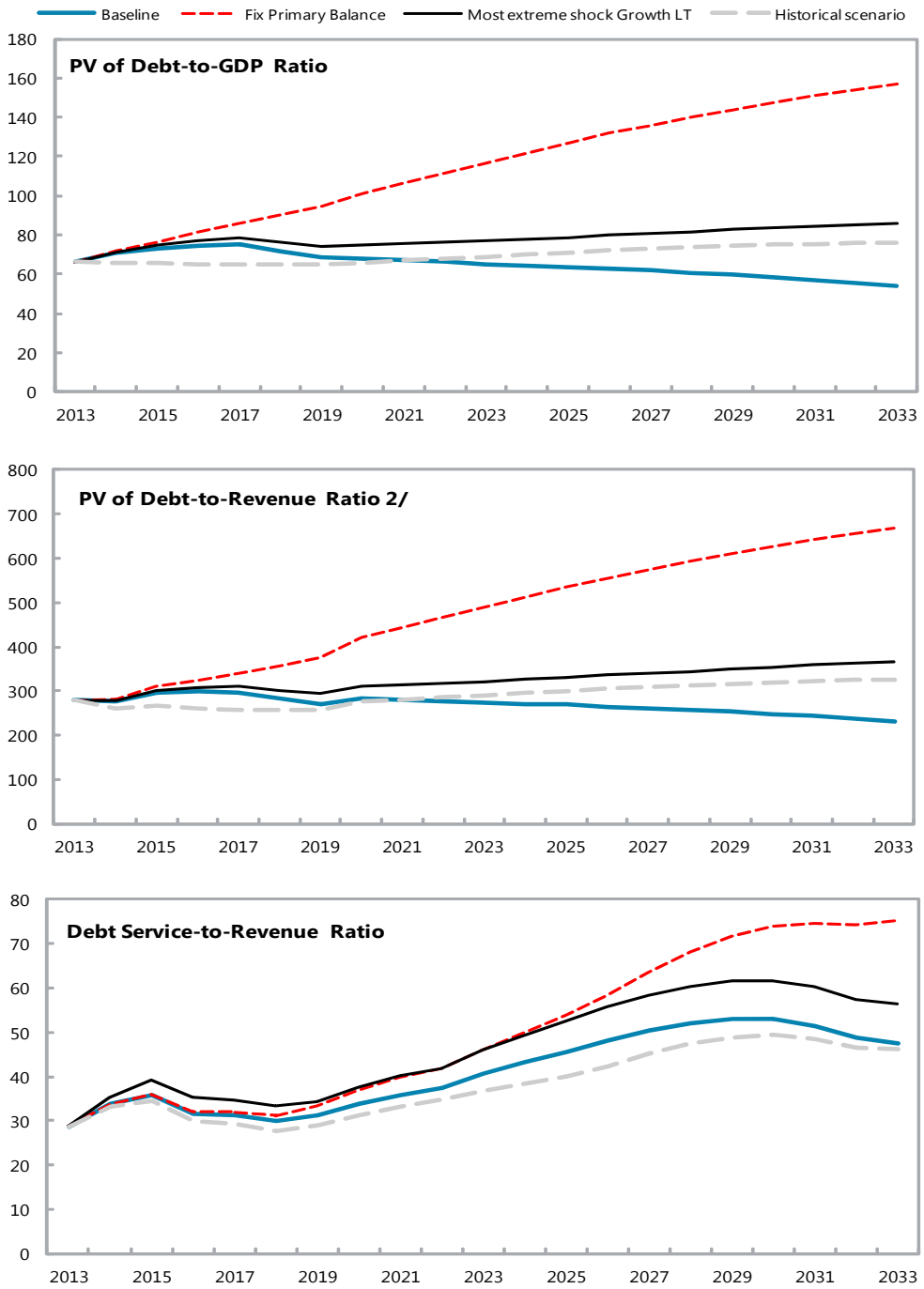
Figure 1. Cabo Verde: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2013–33 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Cabo Verde: Indicators of Public Debt Under Alternative Scenarios, 2013–33 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023.

2/ Revenues are defined inclusive of grants.

Table 1a. Cabo Verde: External Debt Sustainability Framework, Baseline Scenario, 2010–33 1/

(Percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**70.7**	**81.4**	**99.8**			**110.1**	**110.8**	**111.3**	**109.2**	**105.8**	**100.0**		**80.2**	**48.8**	
of which: public and publicly guaranteed (PPG)	51.3	56.1	67.5			74.0	79.2	81.9	82.0	80.5	76.5		62.5	37.7	
Change in external debt	6.8	10.7	18.4			10.3	0.7	0.4	-2.0	-3.4	-5.8		-4.0	-1.9	
Identified net debt-creating flows	7.5	3.1	12.1			2.0	3.3	4.1	3.5	1.9	0.1		-1.4	-3.2	
Non-interest current account deficit	**8.5**	**13.3**	**8.6**	**8.9**	**4.1**	**2.8**	**7.1**	**8.7**	**8.5**	**6.9**	**5.0**		**2.7**	**-0.1**	1.8
Deficit in balance of goods and services	28.4	31.6	21.9			14.7	17.7	18.8	18.0	15.7	13.3		8.1	0.5	
Exports	38.4	42.3	43.9			44.8	45.3	46.5	47.6	48.8	50.4		52.6	53.0	
Imports	66.9	73.9	65.8			59.4	63.0	65.3	65.6	64.6	63.7		60.7	53.5	
Net current transfers (negative = inflow)	-20.4	-19.2	-14.6	-20.6	3.4	-13.5	-12.3	-11.8	-11.2	-10.8	-10.3		-7.9	-4.4	-6.8
of which: official	-6.3	-3.7	-3.3			-3.1	-1.6	-1.5	-1.4	-1.3	-1.2		-0.9	-0.5	
Other current account flows (negative = net inflow)	0.5	1.0	1.2			1.6	1.6	1.7	1.8	1.9	2.0		2.5	3.8	
Net FDI (negative = inflow)	**-6.7**	**-5.6**	**-3.3**	**-7.3**	**3.0**	**-1.9**	**-2.6**	**-3.1**	**-3.2**	**-3.3**	**-3.2**		**-3.1**	**-3.1**	-3.1
Endogenous debt dynamics 2/	**5.6**	**-4.6**	**6.8**			**1.1**	**-1.1**	**-1.5**	**-1.8**	**-1.7**	**-1.6**		**-1.1**	**0.0**	
Contribution from nominal interest rate	3.8	3.0	2.8			1.6	2.0	2.1	2.3	2.3	2.2		2.1	1.9	
Contribution from real GDP growth	-1.0	-2.5	-1.0			-0.5	-3.1	-3.6	-4.1	-4.0	-3.9		-3.2	-1.9	
Contribution from price and exchange rate changes	2.8	-5.0	5.1			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-0.7**	**7.5**	**6.3**			**8.3**	**-2.6**	**-3.6**	**-5.5**	**-5.3**	**-5.9**		**-2.6**	**1.2**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	70.5			78.7	77.5	77.2	75.7	73.6	70.2		60.2	39.3	
In percent of exports	...	...	160.5			175.7	171.0	166.0	159.0	150.7	139.3		114.6	74.1	
PV of PPG external debt	**...**	**...**	**38.2**			**42.6**	**45.8**	**47.8**	**48.5**	**48.2**	**46.7**		**42.6**	**28.2**	
In percent of exports	**...**	**...**	**87.0**			**95.0**	**101.2**	**102.9**	**101.8**	**98.8**	**92.7**		**80.9**	**53.1**	
In percent of government revenues	**...**	**...**	**180.6**			**199.5**	**203.0**	**208.7**	**205.9**	**199.9**	**193.7**		**185.4**	**122.4**	
Debt service-to-exports ratio (in percent)	**13.5**	**9.8**	**8.8**			**6.1**	**7.1**	**7.2**	**7.4**	**7.1**	**6.8**		**8.3**	**9.1**	
PPG debt service-to-exports ratio (in percent)	**4.7**	**4.2**	**4.3**			**4.2**	**4.5**	**4.5**	**4.4**	**4.1**	**4.0**		**5.8**	**7.0**	
PPG debt service-to-revenue ratio (in percent)	**8.4**	**7.8**	**8.9**			**8.9**	**9.0**	**9.1**	**8.8**	**8.4**	**8.3**		**13.2**	**16.0**	
Total gross financing need (Billions of U.S. dollars)	0.1	0.2	0.2			0.1	0.2	0.2	0.2	0.2	0.1		0.2	0.1	
Non-interest current account deficit that stabilizes debt ratio	1.8	2.7	-9.8			-7.5	6.3	8.3	10.6	10.3	10.8		6.8	1.8	
Key macroeconomic assumptions															
Real GDP growth (in percent)	1.5	4.0	1.2	4.9	3.5	0.5	3.0	3.5	4.0	4.0	4.0	3.2	4.0	4.0	3.9
GDP deflator in US dollar terms (change in percent)	-4.3	7.7	-5.8	5.4	8.8	5.9	5.6	4.4	3.8	4.1	4.0	4.6	2.5	2.5	2.5
Effective interest rate (percent) 5/	5.7	4.7	3.3	3.7	1.4	1.7	1.9	2.1	2.2	2.3	2.3	2.1	2.6	4.1	3.0
Growth of exports of G&S (US dollar terms, in percent)	12.2	23.0	-1.0	15.6	15.9	8.6	10.1	10.8	10.6	11.0	11.6	10.4	6.6	6.7	6.9
Growth of imports of G&S (US dollar terms, in percent)	2.3	23.6	-15.1	11.8	16.4	-3.8	15.4	11.9	8.5	6.6	6.6	7.5	5.2	5.3	5.3
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	42.6	46.0	45.7	45.0	44.9	43.9	44.7	28.4	18.2	23.8
Government revenues (excluding grants, in percent of GDP)	21.7	22.7	21.2			21.3	22.6	22.9	23.5	24.1	24.1		22.9	23.0	23.0
Aid flows (in Billions of US dollars) 7/	0.3	0.3	0.3			0.3	0.3	0.3	0.2	0.2	0.1		0.1	0.0	
of which: Grants	0.1	0.1	0.0			0.0	0.1	0.0	0.0	0.0	0.0		0.0	0.0	
of which: Concessional loans	0.2	0.2	0.2			0.2	0.2	0.2	0.2	0.2	0.1		0.1	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			7.8	8.1	6.3	5.1	4.0	2.6		1.8	1.2	1.6
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			51.6	57.2	54.0	54.1	54.5	59.3		43.4	27.1	37.2
Memorandum items:															
Nominal GDP (Billions of US dollars)	1.7	1.9	1.8			1.9	2.1	2.2	2.4	2.6	2.8		3.8	7.2	
Nominal dollar GDP growth	-2.8	11.9	-4.7			6.5	8.8	8.0	8.0	8.3	8.1	7.9	6.6	6.6	6.5
PV of PPG external debt (in Billions of US dollars)			0.7			0.8	1.0	1.1	1.2	1.3	1.3		1.7	2.1	
(PVt-PVt-1)/GDPt-1 (in percent)						**7.8**	**6.6**	**5.9**	4.9	3.9	2.2	**5.2**	1.6	0.9	1.3
Gross workers' remittances (Billions of US dollars)	0.1	0.2	0.2			0.2	0.2	0.2	0.2	0.2	0.2		0.2	0.2	
PV of PPG external debt (in percent of GDP + remittances)	...	...	34.9			39.1	42.3	44.2	44.9	44.8	43.5		40.3	27.3	
PV of PPG external debt (in percent of exports + remittances)	...	...	71.6			79.5	85.3	87.5	87.3	85.5	80.9		73.0	50.2	
Debt service of PPG external debt (in percent of exports + remittance	...	...	3.5			3.5	3.8	3.8	3.7	3.6	3.5		5.2	6.6	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b. Cabo Verde: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–33

(Percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio								
Baseline	43	46	48	48	48	47	**43**	28
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	43	44	42	40	38	35	**32**	33
A2. New public sector loans on less favorable terms in 2013-2033 2	43	49	54	56	57	56	**54**	43
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	43	47	50	51	51	49	**46**	31
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	43	50	58	58	58	56	**51**	30
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	43	51	57	58	58	56	**52**	35
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	43	42	39	40	41	39	**38**	29
B5. Combination of B1-B4 using one-half standard deviation shocks	43	43	43	44	44	43	**41**	31
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	43	65	68	69	69	67	**61**	41
PV of debt-to-exports ratio								
Baseline	95	101	103	102	99	93	**81**	53
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	95	97	91	83	77	70	**62**	63
A2. New public sector loans on less favorable terms in 2013-2033 2	95	109	115	118	117	111	**103**	82
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	95	102	104	104	101	95	**84**	55
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	95	121	152	150	145	136	**118**	70
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	95	102	104	104	101	95	**84**	55
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	95	92	84	85	83	78	**71**	54
B5. Combination of B1-B4 using one-half standard deviation shocks	95	92	89	89	88	82	**75**	57
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	95	102	104	104	101	95	**84**	55
PV of debt-to-revenue ratio								
Baseline	199	203	209	206	200	194	**185**	122
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	199	194	184	168	156	147	**141**	145
A2. New public sector loans on less favorable terms in 2013-2033 2	199	219	234	238	237	232	**236**	189
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	199	209	220	218	212	205	**199**	133
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	199	219	251	247	239	232	**220**	130
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	199	225	250	248	241	233	**227**	151
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	199	185	171	172	169	163	**164**	125
B5. Combination of B1-B4 using one-half standard deviation shocks	199	189	186	186	183	177	**178**	135
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	199	287	295	293	285	276	**268**	178

Table 1b. Cabo Verde: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–33 (continued)

(Percent)

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
Debt service-to-exports ratio								
Baseline	4	5	4	4	4	4	**6**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	4	4	4	4	3	3	**4**	5
A2. New public sector loans on less favorable terms in 2013-2033 2	4	5	5	5	5	5	**7**	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	4	5	4	4	4	4	**6**	7
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	4	5	6	6	6	5	**9**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	4	5	4	4	4	4	**6**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	4	5	4	4	4	4	**4**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	4	5	4	4	4	4	**4**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	4	5	4	4	4	4	**6**	7
Debt service-to-revenue ratio								
Baseline	9	9	9	9	8	8	**13**	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	9	9	9	8	7	7	**9**	11
A2. New public sector loans on less favorable terms in 2013-2033 2	9	9	9	10	10	10	**16**	23
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	9	9	9	9	9	9	**14**	17
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	9	9	9	10	9	9	**17**	18
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	9	10	11	10	10	10	**16**	19
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	9	9	9	8	7	7	**10**	15
B5. Combination of B1-B4 using one-half standard deviation shocks	9	9	9	9	8	8	**10**	16
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	9	13	13	12	12	12	**18**	23
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	29	29	29	29	29	29	**29**	29

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Cabo Verde: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010–33

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	73.0	77.8	90.2			97.6	104.1	107.4	108.4	107.5	101.4		85.2	63.5	
of which: foreign-currency denominated	51.3	56.1	67.5			74.0	79.2	81.9	82.0	80.5	76.5		62.5	37.7	
Change in public sector debt	8.4	4.8	12.4			7.4	6.5	3.3	1.0	-0.9	-6.1		-2.6	-2.1	
Identified debt-creating flows	14.4	6.8	11.5			7.7	5.8	3.1	0.8	-1.1	-6.4		-2.9	-2.4	
Primary deficit	9.8	9.7	11.7	4.4	4.6	11.2	9.5	8.2	6.6	4.6	-0.4	6.6	1.0	-0.1	0.6
Revenue and grants	27.9	25.6	23.0			23.7	25.5	24.7	25.1	25.4	25.3		23.8	23.5	
of which: grants	6.3	2.9	1.8			2.4	2.9	1.8	1.6	1.3	1.2		0.9	0.5	
Primary (noninterest) expenditure	37.7	35.3	34.7			34.9	35.0	32.9	31.7	30.1	24.9		24.8	23.4	
Automatic debt dynamics	4.8	-2.9	-0.3			-3.5	-3.7	-5.0	-5.8	-5.7	-6.0		-3.9	-2.3	
Contribution from interest rate/growth differential	0.0	-2.7	-0.4			0.0	-2.3	-3.2	-3.9	-4.0	-4.6		-3.6	-2.2	
of which: contribution from average real interest rate	0.9	0.1	0.5			0.5	0.6	0.3	0.2	0.1	-0.6		-0.2	0.4	
of which: contribution from real GDP growth	-0.9	-2.8	-0.9			-0.5	-2.9	-3.6	-4.1	-4.1	-4.1		-3.4	-2.5	
Contribution from real exchange rate depreciation	4.8	-0.2	0.2			-3.5	-1.4	-1.8	-1.9	-1.7	-1.3		...	...	
Other identified debt-creating flows	-0.2	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.2	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-6.0	-2.0	0.9			-0.2	0.7	0.2	0.2	0.2	0.3		0.3	0.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	60.9			66.2	70.7	73.3	74.8	75.3	71.6		65.2	54.0	
of which: foreign-currency denominated	...	...	38.2			42.6	45.8	47.8	48.5	48.2	46.7		42.6	28.2	
of which: external	...	...	38.2			42.6	45.8	47.8	48.5	48.2	46.7		42.6	28.2	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	15.0	15.0	17.8			18.0	18.1	17.0	14.5	12.6	7.2		10.7	11.1	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	264.9			279.3	277.0	296.8	298.1	296.1	282.8		273.6	229.9	
PV of public sector debt-to-revenue ratio (in percent)	...	...	287.6			310.2	313.2	320.0	318.0	311.9	297.0		284.2	234.6	
of which: external 3/	...	...	180.6			199.5	203.0	208.7	205.9	199.9	193.7		185.4	122.4	
Debt service-to-revenue and grants ratio (in percent) 4/	18.6	20.5	26.5			28.6	33.7	35.7	31.7	31.3	29.8		40.6	47.5	
Debt service-to-revenue ratio (in percent) 4/	23.9	23.2	28.8			31.8	38.1	38.5	33.8	32.9	31.3		42.2	48.4	
Primary deficit that stabilizes the debt-to-GDP ratio	1.4	4.9	-0.7			3.8	3.0	4.9	5.6	5.5	5.7		3.6	2.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	1.5	4.0	1.2	4.9	3.5	0.5	3.0	3.5	4.0	4.0	4.0	3.2	4.0	4.0	3.9
Average nominal interest rate on forex debt (in percent)	1.1	1.3	1.4	1.2	0.1	1.2	1.2	1.2	1.1	1.1	1.1	1.2	1.2	2.2	1.5
Average real interest rate on domestic debt (in percent)	4.6	2.0	3.4	3.1	1.9	2.8	3.6	3.4	3.5	3.2	0.7	2.8	1.3	1.3	1.3
Real exchange rate depreciation (in percent, + indicates depreciation	11.2	-0.4	0.3	-1.4	10.8	-5.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	0.5	2.7	1.9	1.8	1.3	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Growth of real primary spending (deflated by GDP deflator, in percer	0.2	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	-0.1	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	42.6	46.0	45.7	45.0	44.9	43.9	44.7	28.4	18.2	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b. Cabo Verde: Sensitivity Analysis for Key Indicators of Public Debt, 2013–33

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	66	71	73	75	75	72	65	54
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	66	66	65	65	65	65	69	76
A2. Primary balance is unchanged from 2013	66	72	77	81	86	90	117	157
A3. Permanently lower GDP growth 1/	66	71	75	77	79	76	77	86
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-20	66	72	77	79	80	77	73	65
B2. Primary balance is at historical average minus one standard deviations in 2014-201	66	70	74	75	75	72	65	54
B3. Combination of B1-B2 using one half standard deviation shocks	66	69	71	72	73	69	64	54
B4. One-time 30 percent real depreciation in 2014	66	86	86	86	85	81	73	60
B5. 10 percent of GDP increase in other debt-creating flows in 2014	66	78	80	81	81	78	70	57
PV of Debt-to-Revenue Ratio 2/								
Baseline	279	277	297	298	296	283	274	230
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	279	259	265	261	258	257	290	325
A2. Primary balance is unchanged from 2013	279	282	310	324	338	356	491	669
A3. Permanently lower GDP growth 1/	279	279	302	307	309	300	321	365
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-20	279	282	311	315	315	304	306	278
B2. Primary balance is at historical average minus one standard deviations in 2014-201	279	276	298	299	297	284	274	230
B3. Combination of B1-B2 using one half standard deviation shocks	279	269	286	288	287	274	267	228
B4. One-time 30 percent real depreciation in 2014	279	338	350	343	335	319	307	254
B5. 10 percent of GDP increase in other debt-creating flows in 2014	279	305	324	324	321	307	296	243
Debt Service-to-Revenue Ratio 2/								
Baseline	29	34	36	32	31	30	41	47
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	29	33	34	30	29	28	37	46
A2. Primary balance is unchanged from 2013	29	34	36	32	32	31	46	75
A3. Permanently lower GDP growth 1/	29	34	36	32	32	31	44	60
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-20	29	34	37	33	33	31	43	52
B2. Primary balance is at historical average minus one standard deviations in 2014-201	29	34	36	32	31	30	41	48
B3. Combination of B1-B2 using one half standard deviation shocks	29	34	36	31	31	30	40	47
B4. One-time 30 percent real depreciation in 2014	29	35	39	35	35	33	46	57
B5. 10 percent of GDP increase in other debt-creating flows in 2014	29	34	36	33	32	30	43	49

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.